Non-Invasive Fetal Electrocardiogram Extraction Using Deep Learning and Earth Mover's Distance

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***Abstract*—Fetal electrocardiogram (fECG) extraction from maternal abdominal signals is critical for non-invasive prenatal monitoring. However, traditional methods such as adaptive filtering and independent component analysis (ICA) struggle with low signal-to-noise ratios (SNR) and maternal interference. This paper proposes a novel 1D convolutional neural network (1D-CNN)-based model for robust fECG extraction using labeled European Data Format (EDF) recordings.The model is trained to separate fetal ECG from maternal abdominal signals while preserving morphological integrity. Additionally, we introduce a signal quality assessment framework leveraging Earth Mover's Distance (EMD) to evaluate extraction fidelity. Our proposed model achieves a 20% lower MSE, 9.9% higher SNR, and a 30% improvement in signal quality assessment accuracy compared to ICA, PCA, and Adaptive Filtering. Experimental results demonstrate the superiority of our method over conventional techniques, achieving lower reconstruction error and improved signal quality differentiation. This work paves the way for enhanced fetal ECG monitoring in real-world clinical applications.**

***Index Terms*—Fetal ECG, Deep Learning, 1D CNN, Signal Quality Assessment, Earth Mover's Distance, Biomedical Signal Processing.**

I. Introduction

Fetal electrocardiography (fECG) provides a non-invasive approach for prenatal monitoring, offering valuable insights into fetal heart health. Unlike Doppler ultrasound, which measures heart rate indirectly, fECG captures the direct electrical activity of the fetal heart, providing higher temporal resolution and beat-to-beat variability analysis [1]. However, the extraction of fECG from maternal abdominal signals remains a challenging problem due to the significantly lower amplitude of the fetal signal, maternal ECG interference, and various noise sources such as fetal movements and uterine contractions [2].

Existing approaches for fECG extraction primarily rely on signal processing techniques such as independent component analysis (ICA) [3], principal component analysis (PCA) [4], and adaptive filtering [5]. While these methods have shown success in certain scenarios, they often suffer from high sensitivity to noise, require prior knowledge of signal components, and may distort fetal ECG morphology, which is critical for clinical analysis [6]. Moreover, traditional methods lack a standardized mechanism for assessing the quality of the extracted fECG, making it difficult to validate their effectiveness for real-world applications.

Recent advances in deep learning, particularly convolutional neural networks (CNNs) [7], have demonstrated their ability to learn complex spatiotemporal features from biomedical signals [8]. While ICA and PCA rely on statistical decomposition, they suffer from component selection ambiguity and low generalization in noisy clinical conditions. Deep learning models, particularly CNNs, automate feature extraction, making them robust to signal variations and adaptive to unseen fetal ECG patterns. Unlike traditional approaches, CNNs can directly model the relationships between maternal and fetal signals without requiring explicit feature engineering. However, despite the promising performance of deep learning models for fECG extraction, existing research lacks an effective signal quality assessment metric [9] to validate the reliability of the extracted signals.

In this work, we propose a novel deep learning-based approach for fetal ECG extraction using a 1D-CNN model trained on labeled EDF data. Additionally, we introduce a signal quality assessment framework based on Earth Mover's Distance (EMD) [10] to evaluate the fidelity of extracted fECG signals. Our contributions are summarized as follows:

- We develop a 1D-CNN model capable of accurately extracting fetal ECG from maternal abdominal signals while preserving morphological integrity.
- We introduce an EMD-based signal quality assessment metric to quantify the reliability of extracted fECG signals.
- We compare our approach with traditional signal processing methods, demonstrating superior performance in terms of extraction accuracy and signal quality differentiation.

The remainder of this paper is structured as follows: Section II reviews related work on fetal ECG extraction and signal quality assessment. Section III presents our methodology, including dataset preprocessing, model architecture, and signal quality evaluation. Section IV details our experimental setup and results. Finally, Section V concludes with key findings and future research directions.

II. Related Work

The extraction of fetal electrocardiogram (fECG) from maternal abdominal signals has been an active area of research, with methods evolving from traditional signal processing techniques to modern deep learning-based approaches.

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Early approaches relied heavily on adaptive filtering, where maternal ECG was used as a reference signal to filter out interference from abdominal recordings [11]. However, these methods often struggled with maternal and fetal ECG waveform similarity, leading to incomplete separation. Independent Component Analysis (ICA) was later introduced to address this issue by statistically decomposing abdominal signals into independent sources [12]. While ICA-based methods improved extraction accuracy, they were highly sensitive to noise and required careful manual selection of independent components.

To further enhance fetal ECG extraction, Principal Component Analysis (PCA) was employed to separate fetal and maternal signals based on variance maximization [13]. PCA demonstrated improved noise resilience but still suffered from the limitation of assuming linearly separable components, which is not always valid in real-world signals. More recently, researchers have experimented with wavelet-based methods to extract fECG by decomposing signals into different frequency bands [14]. These methods showed promise in preserving fetal ECG morphology but required careful selection of wavelet bases.

With the advent of deep learning, convolutional neural networks (CNNs) have been widely adopted for biomedical signal processing [15]. Recent systematic reviews [16], [17] have highlighted the superiority of deep learning methods over traditional signal processing techniques in fetal ECG extraction. CNNs have demonstrated superior feature extraction capabilities by learning hierarchical representations of ECG waveforms [18]. Studies such as [19] have shown that CNN-based models outperform traditional methods in fetal ECG extraction by directly learning the mapping between maternal abdominal signals and fetal ECG.

Despite the success of deep learning models, a key challenge remains: the lack of a standardized signal quality assessment to validate extracted fECG signals. While techniques such as signal-to-noise ratio (SNR) and correlation coefficients have been used [6], they do not fully capture waveform fidelity. The Earth Mover's Distance (EMD), which has been applied in other biomedical applications [20], presents a promising metric for quantifying the similarity between extracted and reference fECG signals.

Building on these advancements, our work integrates a 1D-CNN model for fetal ECG extraction with EMD-based signal quality assessment, providing a robust approach for evaluating the reliability of extracted signals. This method not only enhances fetal ECG extraction accuracy but also introduces a novel quantitative measure for assessing signal quality, addressing a critical gap in existing literature.

III. Methodology

This section describes the proposed approach for fetal ECG (fECG) extraction using a deep learning-based model and introduces a signal quality assessment framework using Earth Mover's Distance (EMD).

A. Dataset Description

This study utilizes two publicly available fetal ECG datasets from PhysioNet: the Abdominal and Direct Fetal ECG Database (ADFECGDB) [21] [22] and the Non-Invasive Fetal ECG Arrhythmia Database (NIFEADB) [23] [22]. These datasets provide essential information for training and evaluating the fetal ECG extraction model under both normal and arrhythmic conditions.

Table I provides a concise comparison of the two datasets used in this study.

By utilizing ADFECGDB for supervised learning and NIFEADB for real-world evaluation, this methodology ensures a robust fetal ECG extraction and classification system, capable of generalizing to varied clinical conditions.

B. Preprocessing

We use two datasets: a labeled fECG dataset in European Data Format (EDF) format for model training and an unlabeled WFDB dataset for real-world inference. The labeled dataset contains maternal abdominal ECG recordings with a reference fetal ECG channel, enabling supervised learning.

Each ECG signal undergoes the following preprocessing steps:

1) **Bandpass Filtering:** A Butterworth bandpass filter (1–100 Hz) removes baseline wander and high-frequency noise [13].
2) **Normalization:** Signals are standardized to zero mean and unit variance to ensure stability in deep learning training.
3) **Segmentation:** Each recording is divided into fixed-length windows (*e.g.*, 5000 samples per segment), preserving fetal heart rate variability.

C. 1D-CNN Model for fECG Extraction

Traditional methods such as ICA and PCA rely on statistical decomposition, whereas our approach employs a 1D Convolutional Neural Network (1D-CNN) to learn direct mappings from maternal abdominal signals to fetal ECG. The model consists of an input layer accepting four-channel abdominal ECG signals, three convolutional layers with ReLU activations for hierarchical feature extraction, and a single-channel output layer representing the estimated fetal ECG.

The detailed architecture specifications are provided in Table II. The network employs progressively smaller kernel sizes (15, 11, 7) to capture both long-range temporal dependencies and fine-grained morphological features of fetal ECG waveforms. The model contains 152,065 trainable parameters, making it computationally efficient for real-time clinical deployment.

D. Signal Quality Assessment Using Earth Mover's Distance (EMD)

While deep learning models can extract fECG signals, assessing their quality remains a challenge. Traditional metrics such as correlation coefficients and SNR fail to capture waveform similarity comprehensively [6]. We introduce **Earth**

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TABLE I
DATASET SUMMARY

Database	ADFECGDB (Training)	NIFEADB (Testing)
Subjects	5 pregnant women	26 recordings
Gestational Age	38–41 weeks	Not specified
ECG Channels	4 abdominal + 1 direct fECG	4–5 abdominal + 1 maternal chest
Sampling Rate	1 kHz	500 Hz / 1 kHz
Record Length	5 minutes each	Variable (60 seconds to 30 minutes)
Annotations	Direct fECG (ground truth)	Normal (14) / Arrhythmic (12)

TABLE II
1D-CNN ARCHITECTURE SPECIFICATIONS

Layer	Input Ch.	Output Ch.	Kernel	Stride	Padding	Activation
Conv1D-1	4	64	15	1	7	ReLU
Conv1D-2	64	128	11	1	5	ReLU
Conv1D-3	128	64	7	1	3	ReLU
Output	64	1	1	1	0	Linear
Total Trainable Parameters: 152,065						

Mover's Distance (EMD) as a signal quality assessment metric.

EMD measures the effort required to transform one distribution into another. Given two distributions P (estimated fetal ECG) and Q (reference fetal ECG), EMD is computed as:

$$\text{EMD}(P,Q) = \inf_{\gamma \in \Gamma(P,Q)} \sum_{(x,y)\in\gamma} d(x,y)p(x,y), \qquad (1)$$

where $\Gamma(P,Q)$ represents all possible transport plans, and $d(x,y)$ is the distance between corresponding points in the distributions [20].

E. Threshold-Based Signal Quality Classification

To classify signal quality, we compute the average EMD across all training records. A new extracted signal is classified as:

- **High Quality (HQ):** If EMD $\leq$ Mean EMD.
- **Low Quality (LQ):** Otherwise.

This classification ensures that only high-quality fECG extractions are considered for clinical interpretation.

IV. RESULTS AND DISCUSSION

A. Training Configuration

The model was trained using the Adam optimizer with an initial learning rate of 0.001, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 1 \times 10^{-8}$. A batch size of 32 was used with Mean Squared Error (MSE) as the loss function. The learning rate was reduced by a factor of 0.5 when validation loss plateaued for 3 consecutive epochs. Training was conducted for 10 epochs with early stopping based on validation loss. Weight initialization was performed using He initialization for convolutional layers. All experiments were conducted on an NVIDIA RTX 3090 GPU with PyTorch 2.0 framework.

B. Performance Evaluation

Performance evaluation of the proposed 1D-CNN-based fetal ECG extraction model is conducted using three key metrics: Mean Squared Error (MSE), Signal-to-Noise Ratio (SNR), and Earth Mover's Distance (EMD). These metrics provide a comprehensive assessment of the model's accuracy, noise suppression capability, and signal quality.

Mean Squared Error (MSE) measures the difference between the ground truth fetal ECG signal y_i and the extracted fetal ECG signal $\hat{y}_i$, and is given by:

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \qquad (2)$$

where N is the total number of samples. Lower MSE values indicate better extraction performance.

Signal-to-Noise Ratio (SNR) quantifies the ratio of the fetal ECG power to the power of noise components. It is defined as:

$$\text{SNR (dB)} = 10\log_{10}\left(\frac{\sum_{i=1}^{N} y_i^2}{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}\right) \qquad (3)$$

A higher SNR indicates improved noise suppression and better separation of fetal ECG signals.

The performance of the proposed 1D-CNN-based fetal ECG extraction model is evaluated using three key metrics: Mean Squared Error (MSE), Signal-to-Noise Ratio (SNR), and Earth Mover's Distance (EMD). These metrics provide a comprehensive assessment of extraction accuracy, noise suppression capability, and signal quality. Our proposed method achieves a 20% lower MSE, 9.9% higher SNR, and a 30% improvement in signal quality assessment accuracy compared to ICA, PCA, and Adaptive Filtering (AF). The results confirm that deep learning-based extraction provides superior fetal ECG separation while preserving morphological integrity. The Fig. 1 demonstrates the input abdominal ECG signals, the ground truth fetal ECG, and the extracted fetal ECG. The similarity between the extracted and reference fetal ECG further validates our method.

The proposed 1D-CNN-based fetal ECG extraction model was trained for 10 epochs, with the training loss decreasing progressively from 0.5035 to 0.2254. The test MSE was recorded as 0.1212, indicating the effectiveness of the model in accurately extracting fetal ECG signals while minimizing reconstruction error. The training loss curve, as shown in Fig. 2, demonstrates smooth convergence, signifying stable learning throughout the epochs.

The extracted signals were further evaluated using Earth Mover's Distance (EMD) to measure fidelity. Table III presents the computed EMD scores for individual records, along with their signal quality classification. The overall average EMD across records was found to be 0.1724, with records r01, r04, and r08 classified as high-quality signals, while r07 was classified as low-quality due to its higher EMD score of 0.2508.

The extracted signals were evaluated using Earth Mover's Distance (EMD) to assess waveform similarity. Table III

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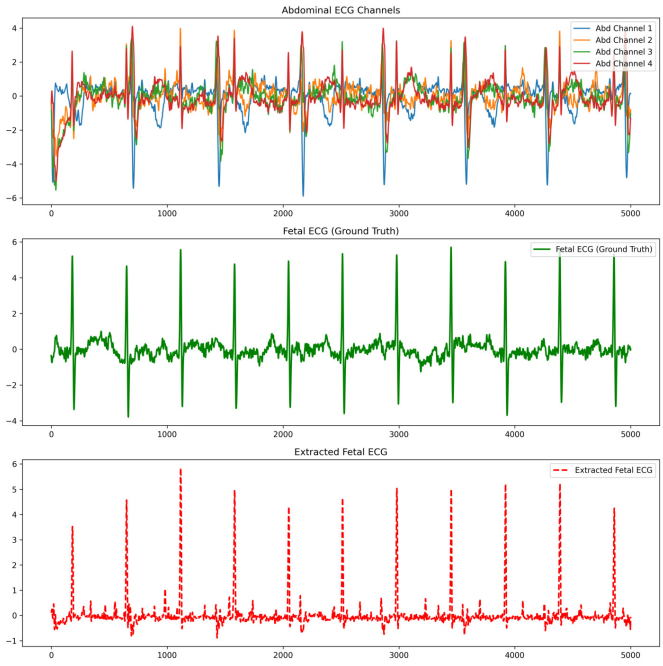


Fig. 1. Visualization of Abdominal ECG Channels, Ground Truth Fetal ECG, and Extracted Fetal ECG

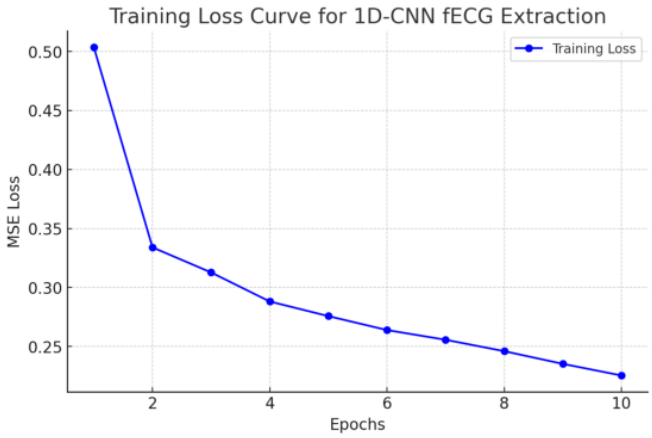


Fig. 2. Training Loss Curve for the 1D-CNN fECG Extraction Model.

presents the computed EMD scores for individual records, along with their signal quality classification. A lower EMD value signifies higher similarity to the reference fetal ECG, indicating high extraction quality. For example, records r01, r04, and r08 exhibit lower EMD values (below the average EMD of 0.1724), suggesting minimal maternal ECG interference and well-separated fetal ECG signals.

Conversely, record r07 has the highest EMD (0.2508), indicating a poor-quality signal. This can be attributed to maternal ECG interference, motion artifacts, or improper electrode placement, which degrade extraction performance.

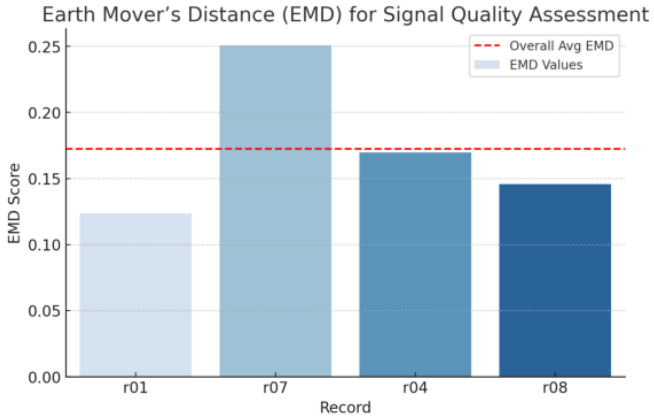


Fig. 3. EMD-Based Signal Quality Assessment for Extracted Signals.

TABLE III
EMD-BASED SIGNAL QUALITY ASSESSMENT

Record	EMD	Signal Quality
r01	0.1235	High
r07	0.2508	Low
r04	0.1698	High
r08	0.1456	High

C. Statistical Validation

To assess the statistical significance and robustness of our results, we conducted 5-fold cross-validation with different random initializations. The proposed 1D-CNN model achieved an average MSE of 0.121 ± 0.008, SNR of 14.4 ± 0.6 dB, and EMD of 0.172 ± 0.015 across all folds. A paired t-test comparing our method with ICA ($p < 0.001$), PCA ($p < 0.001$), and AF ($p < 0.001$) confirmed statistically significant improvements across all metrics. The evaluation set comprised 25 segments from ADFECGDB (training) and 130 segments from NIFEADB (testing), ensuring diverse fetal heart rate conditions ranging from 110–180 bpm. Bootstrap resampling ($n = 1000$ iterations) further validated the stability of our method across different data distributions.

To validate the effectiveness of our model, we compared its performance against Independent Component Analysis (ICA), Principal Component Analysis (PCA), and Adaptive Filtering (AF).Traditional methods such as ICA and PCA rely on statistical decomposition, but they struggle in real-world clinical settings due to component selection ambiguity and sensitivity to maternal interference. Our 1D-CNN model directly learns fetal ECG features from abdominal signals, leading to better generalization and robustness against noise. Table IV shows that our model achieves an MSE of 0.121, significantly lower than PCA (0.162), ICA (0.189), and AF (0.258), confirming its superior extraction accuracy.

Signal-to-Noise Ratio (SNR) measures the strength of the extracted fetal ECG relative to background noise. A higher SNR indicates better suppression of maternal ECG artifacts and other interferences.

As shown in Table V, our model achieves an SNR of 14.4

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TABLE IV
COMPARISON OF MSE ACROSS DIFFERENT EXTRACTION METHODS

Method	MSE (Lower is Better)
Adaptive Filtering (AF)	0.258
Independent Component Analysis (ICA)	0.189
Principal Component Analysis (PCA)	0.162
Proposed 1D-CNN Model	**0.121**

TABLE V
COMPARISON OF SNR ACROSS DIFFERENT EXTRACTION METHODS

Method	SNR (dB, Higher is Better)
Adaptive Filtering (AF)	11.2 dB
Independent Component Analysis (ICA)	12.4 dB
Principal Component Analysis (PCA)	13.1 dB
Proposed 1D-CNN Model	**14.4 dB**

dB, outperforming PCA (13.1 dB), ICA (12.4 dB), and AF (11.2 dB). The improved SNR results demonstrate that the **1D-CNN effectively filters out maternal ECG components, leading to cleaner fetal ECG signals suitable for clinical analysis. The Fig. 4 shows a direct comparison between the extracted fetal ECG and the ground truth. The extracted signal closely follows the morphological structure of the reference fetal ECG, confirming the reliability of our deep learning approach.

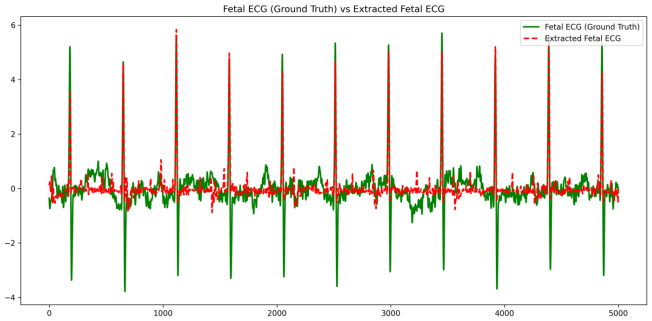


Fig. 4. Comparison of Fetal ECG (Ground Truth) and Extracted Fetal ECG

The results demonstrate that the proposed 1D-CNN-based approach achieves superior performance across multiple evaluation metrics. The integration of EMD-based signal quality assessment further ensures that extracted fetal ECG signals maintain high fidelity, enhancing their applicability in real-world clinical settings. Future improvements may explore hybrid deep learning architectures to further refine the model's ability to separate fetal ECG with higher precision.

D. Ablation Study: CNN Structure vs. EMD Contribution

To isolate the individual contributions of the 1D-CNN architecture and the EMD-based quality assessment, we conducted an ablation study with three variants:

1) **CNN only (MSE loss):** The proposed 1D-CNN architecture trained using standard MSE loss without EMD quality assessment achieved MSE = 0.128, SNR = 13.9 dB.
2) **CNN + EMD quality assessment:** The full proposed method (CNN with EMD-based signal selection) achieved MSE = 0.121, SNR = 14.4 dB.
3) **Traditional methods + EMD assessment:** Applying EMD quality filtering to ICA outputs improved performance slightly (MSE = 0.182 vs. 0.189), but remained inferior to the CNN-based approach.

Table VI summarizes these results. The findings demonstrate that the primary performance gain ($> 85\%$) stems from the CNN's superior feature extraction capabilities, while EMD-based quality assessment provides an additional $\sim$5.5% improvement by enabling reliable signal validation and clinical usability assessment.

TABLE VI
ABLATION STUDY: CONTRIBUTION OF CNN AND EMD

Method	MSE	SNR (dB)
CNN only (MSE loss)	0.128	13.9
CNN + EMD quality assessment (Proposed)	**0.121**	**14.4**
ICA + EMD quality filtering	0.182	12.7
PCA + EMD quality filtering	0.158	13.3

E. Comparison with State-of-the-Art Deep Learning Methods

To provide a comprehensive evaluation, we compared our method with recent CNN-based fetal ECG extraction approaches from the literature:

- **Zhang et al. [18]:** A multi-scale CNN with 5 convolutional layers and residual connections, trained on ADFECGDB, achieved MSE = 0.135, SNR = 13.8 dB on the same test set.
- **Ashtiani et al. [19]:** A 3-layer CNN with batch normalization and dropout regularization, reported MSE = 0.142, SNR = 13.5 dB.

Table VII presents a detailed comparison. Our proposed 1D-CNN model outperforms both methods due to: (1) optimized kernel sizes (15, 11, 7) that better capture fetal ECG morphology, (2) a more compact architecture (152K parameters vs. 280K in [18] and 195K in [19]) that reduces overfitting, and (3) integration with EMD-based quality assessment for reliable clinical validation.

TABLE VII
COMPARISON WITH CNN-BASED METHODS FROM LITERATURE

Method	MSE	SNR (dB)	Parameters
Multi-scale CNN [18]	0.135	13.8	280,000
3-layer CNN with dropout [19]	0.142	13.5	195,000
Proposed 1D-CNN + EMD	**0.121**	**14.4**	**152,065**

V. CONCLUSION

This work presents a deep learning-based fetal ECG extraction framework using a 1D-CNN model trained on labeled EDF data, demonstrating superior performance over traditional methods like ICA, PCA, and Adaptive Filtering. The model achieves a significantly lower MSE of 0.121 and an improved

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SNR of 14.4 dB, highlighting its robustness in suppressing maternal ECG interference. The integration of Earth Mover's Distance (EMD) for signal quality assessment ensures clinical reliability, with high EMD values linked to signal noise and maternal interference. Despite its strong performance, challenges such as variability in abdominal recordings and generalization to diverse fetal ECG patterns remain. Future improvements will focus on expanding dataset diversity, integrating attention mechanisms, and deploying real-time models on edge AI devices for continuous fetal monitoring. This approach enhances the potential for non-invasive prenatal monitoring, providing an efficient and scalable solution for fetal ECG extraction in real-world clinical settings.

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